

ALFAAA13116

## 2,3-Pentanedione

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

**产品说明:** 2,3-戊烷二酮  
**Product Description:** 2,3-Pentanedione

**Cat No. :** A13116  
**Synonyms** Acetylpropionyl  
**CAS No** 600-14-6  
**Molecular Formula** C5 H8 O2

**Supplier** Avocado Research Chemicals Ltd.  
(Part of Thermo Fisher Scientific)  
Shore Road, Heysham  
Lancashire, LA3 2XY,  
United Kingdom  
Office Tel: +44 (0) 1524 850506  
Office Fax: +44 (0) 1524 850608

**Emergency Telephone Number** For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

**E-mail address** begel.sdsdesk@thermofisher.com

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

### SECTION 2. HAZARD IDENTIFICATION

|                                                                                                                                                                                                                                    |                             |                         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-------------------------|
| <b>Physical State</b><br>Liquid                                                                                                                                                                                                    | <b>Appearance</b><br>Yellow | <b>Odor</b><br>Odorless |
| <b>Emergency Overview</b><br>Highly flammable liquid and vapor. May be harmful if swallowed. May cause an allergic skin reaction. Causes serious eye damage.<br>May cause damage to organs through prolonged or repeated exposure. |                             |                         |

#### Classification of the substance or mixture

|                                                      |            |
|------------------------------------------------------|------------|
| Flammable liquids.                                   | Category 2 |
| Acute Oral Toxicity                                  | Category 5 |
| Serious Eye Damage/Eye Irritation                    | Category 1 |
| Skin Sensitization                                   | Category 1 |
| Specific target organ toxicity - (repeated exposure) | Category 2 |

#### Label Elements

**Signal Word****Danger****Hazard Statements**

H225 - Highly flammable liquid and vapor

H303 - May be harmful if swallowed

H317 - May cause an allergic skin reaction

H318 - Causes serious eye damage

H373 - May cause damage to organs through prolonged or repeated exposure

**Precautionary Statements****Prevention**

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

P233 - Keep container tightly closed

P240 - Ground and bond container and receiving equipment

P241 - Use explosion-proof electrical/ ventilating/ lighting equipment

P242 - Use non-sparking tools

P243 - Take action to prevent static discharges

P270 - Do not eat, drink or smoke when using this product

P272 - Contaminated work clothing should not be allowed out of the workplace

P280 - Wear protective gloves/protective clothing/eye protection/face protection

**Response**

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P310 - Immediately call a POISON CENTER or doctor

P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

P362 + P364 - Take off contaminated clothing and wash it before reuse

**Storage**

P403 + P235 - Store in a well-ventilated place. Keep cool

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Physical and Chemical Hazards**

Highly flammable. Vapors may cause flash fire or explosion.

**Health Hazards**

May be harmful if swallowed. May cause an allergic skin reaction. May cause damage to organs through prolonged or repeated exposure.

**Environmental hazards**

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants. Will likely be mobile in the environment due to its water solubility. The product is water soluble, and may spread in water systems.

No information available

This product does not contain any known or suspected endocrine disruptors.

**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

| Component        | CAS No   | Weight % |
|------------------|----------|----------|
| 2,3-Pentanedione | 600-14-6 | <=100    |

**SECTION 4. FIRST AID MEASURES****General Advice**

If symptoms persist, call a physician.

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**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

**Ingestion**

Clean mouth with water and drink afterwards plenty of water.

**Most important symptoms and effects**

None reasonably foreseeable. . Causes severe eye damage. May cause allergic skin reaction. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting: Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing

**Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Notes to Physician**

Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

Carbon dioxide (CO<sub>2</sub>). Dry chemical. Water mist may be used to cool closed containers. Water mist may be used to cool closed containers.

**Extinguishing media which must not be used for safety reasons**

Water may be ineffective.

**Specific Hazards Arising from the Chemical**

Flammable. Vapors may travel to source of ignition and flash back. Containers may explode when heated. Vapors may form explosive mixtures with air.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Ensure adequate ventilation. Use personal protective equipment as required. Remove all sources of ignition. Take precautionary measures against static discharges.

**Environmental Precautions**

Should not be released into the environment.

**Methods for Containment and Clean Up**

Keep in suitable, closed containers for disposal. Soak up with inert absorbent material. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

Refer to protective measures listed in Sections 8 and 13.

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## SECTION 7. HANDLING AND STORAGE

**Handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Take precautionary measures against static discharges.

**Storage**

Keep in a dry place. Keep container tightly closed. Keep away from heat, sparks and flame. Refrigerator/flammables. Keep away from heat/sparks/open flames/hot surfaces. - No smoking. Keep container tightly closed in a dry and well-ventilated place.

**Specific Use(s)**

Use in laboratories

## SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

**Control Parameters**

| Component        | ACGIH TLV | OSHA PEL | NIOSH                        | The United Kingdom | European Union |
|------------------|-----------|----------|------------------------------|--------------------|----------------|
| 2,3-Pentanedione |           |          | TWA: 9.3 ppb<br>STEL: 31 ppb | -                  |                |

Legend

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Monitoring methods**

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS70 General methods for sampling airborne gases and vapours MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography MDHS 96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas chromatography

**Exposure Controls****Engineering Measures**

Use explosion-proof electrical/ventilating/lighting equipment. Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

**Personal protective equipment**

**Eye Protection** Goggles (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Nitrile rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Neoprene       | recommendations   |                 |             |                       |
| Natural rubber |                   |                 |             |                       |
| PVC            |                   |                 |             |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

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|                                        |                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin and body protection</b>        | Wear appropriate protective gloves and clothing to prevent skin exposure                                                                                                                                                                                                                                                    |
| <b>Respiratory Protection</b>          | When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.<br>To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly                                                                                     |
| <b>Large scale/emergency use</b>       | Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced<br><b>Recommended Filter type:</b> Organic gases and vapours filter Type A Brown conforming to EN14387                                                                  |
| <b>Small scale/Laboratory use</b>      | Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.<br><b>Recommended half mask:-</b> Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141<br>When RPE is used a face piece Fit Test should be conducted |
| <b>Hygiene Measures</b>                | Handle in accordance with good industrial hygiene and safety practice.                                                                                                                                                                                                                                                      |
| <b>Environmental exposure controls</b> | No information available.                                                                                                                                                                                                                                                                                                   |

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                                |                                       |                                             |
|------------------------------------------------|---------------------------------------|---------------------------------------------|
| <b>Appearance</b>                              | Yellow                                |                                             |
| <b>Physical State</b>                          | Liquid                                |                                             |
| <b>Odor</b>                                    | Odorless                              |                                             |
| <b>Odor Threshold</b>                          | No data available                     |                                             |
| <b>pH</b>                                      | 4                                     |                                             |
| <b>Melting Point/Range</b>                     | -52 °C / -61.6 °F                     |                                             |
| <b>Softening Point</b>                         | No data available                     |                                             |
| <b>Boiling Point/Range</b>                     | 110 - 112 °C / 230 - 233.6 °F         |                                             |
| <b>Flash Point</b>                             | 18 °C / 64.4 °F                       | <b>Method -</b> No information available    |
| <b>Evaporation Rate</b>                        | No data available                     |                                             |
| <b>Flammability (solid,gas)</b>                | Not applicable                        | Liquid                                      |
| <b>Explosion Limits</b>                        | <b>Lower</b> 1.8<br><b>Upper</b> 10.9 |                                             |
| <b>Vapor Pressure</b>                          | 28.5 hPa @ 20 °C                      |                                             |
| <b>Vapor Density</b>                           | 3.45                                  | (Air = 1.0)                                 |
| <b>Specific Gravity / Density</b>              | 0.958                                 |                                             |
| <b>Bulk Density</b>                            | Not applicable                        | Liquid                                      |
| <b>Water Solubility</b>                        | 60 g/L (15°C)                         |                                             |
| <b>Solubility in other solvents</b>            | No information available              |                                             |
| <b>Partition Coefficient (n-octanol/water)</b> |                                       |                                             |
| <b>Autoignition Temperature</b>                | 265 °C / 509 °F                       |                                             |
| <b>Decomposition Temperature</b>               | > 100°C                               |                                             |
| <b>Viscosity</b>                               | No data available                     |                                             |
| <b>Explosive Properties</b>                    |                                       | Vapors may form explosive mixtures with air |
| <b>Oxidizing Properties</b>                    | No information available              |                                             |
| <b>Molecular Formula</b>                       | C5 H8 O2                              |                                             |
| <b>Molecular Weight</b>                        | 100.12                                |                                             |

## SECTION 10. STABILITY AND REACTIVITY

|                  |                                 |
|------------------|---------------------------------|
| <b>Stability</b> | Stable under normal conditions. |
|------------------|---------------------------------|

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|                                         |                                                                                          |
|-----------------------------------------|------------------------------------------------------------------------------------------|
| <b>Hazardous Reactions</b>              | None under normal processing.                                                            |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                 |
| <b>Conditions to Avoid</b>              | Keep away from open flames, hot surfaces and sources of ignition. Incompatible products. |
| <b>Materials to avoid</b>               | Strong bases. Acids. Bases. Strong oxidizing agents. Metals. Reducing Agent.             |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO). Carbon dioxide (CO <sub>2</sub> ).                                 |

## SECTION 11. TOXICOLOGICAL INFORMATION

## Product Information

## (a) acute toxicity;

| Component        | LD50 Oral             | LD50 Dermal                  | LC50 Inhalation |
|------------------|-----------------------|------------------------------|-----------------|
| 2,3-Pentanedione | LD50 = 3 g/kg ( Rat ) | LD50 > 2000 mg/kg ( Rabbit ) |                 |

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; Category 1

## (d) respiratory or skin sensitization;

## Respiratory

No data available

## Skin

Category 1

May cause sensitization by skin contact

(e) germ cell mutagenicity; No data available

## (f) carcinogenicity;

No data available

There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; Category 2

## Target Organs

No information available.

(j) aspiration hazard; No data available

## Symptoms / effects, both acute and delayed

Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing

## SECTION 12. ECOLOGICAL INFORMATION

Ecotoxicity effects Do not empty into drains. .

## 2,3-Pentanedione

|                                        |                                                                                                                                                           |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Persistence and Degradability</b>   |                                                                                                                                                           |
| <b>Persistence</b>                     | Soluble in water, Persistence is unlikely, based on information available.                                                                                |
| <b>Bioaccumulative Potential</b>       | Bioaccumulation is unlikely                                                                                                                               |
| <b>Mobility in soil</b>                | The product is water soluble, and may spread in water systems Will likely be mobile in the environment due to its water solubility Highly mobile in soils |
| <b>Endocrine Disruptor Information</b> | This product does not contain any known or suspected endocrine disruptors                                                                                 |
| <b>Persistent Organic Pollutant</b>    | This product does not contain any known or suspected substance                                                                                            |
| <b>Ozone Depletion Potential</b>       | This product does not contain any known or suspected substance                                                                                            |

**SECTION 13. DISPOSAL CONSIDERATIONS**

|                                            |                                                                                                                                                                                                                                          |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste from Residues/Unused Products</b> | Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.                                                                   |
| <b>Contaminated Packaging</b>              | Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition. |
| <b>Other Information</b>                   | Waste codes should be assigned by the user based on the application for which the product was used. Do not flush to sewer. Can be landfilled or incinerated, when in compliance with local regulations. Do not empty into drains.        |

**SECTION 14. TRANSPORT INFORMATION****Road and Rail Transport**

|                                |                         |
|--------------------------------|-------------------------|
| <b>UN-No</b>                   | UN1224                  |
| <b>Proper Shipping Name</b>    | KETONES, LIQUID, N.O.S. |
| <b>Technical Shipping Name</b> | (2,3-PENTANEDIONE)      |
| <b>Hazard Class</b>            | 3                       |
| <b>Packing Group</b>           | II                      |

**IMDG/IMO**

|                                |                         |
|--------------------------------|-------------------------|
| <b>UN-No</b>                   | UN1224                  |
| <b>Proper Shipping Name</b>    | KETONES, LIQUID, N.O.S. |
| <b>Technical Shipping Name</b> | (2,3-PENTANEDIONE)      |
| <b>Hazard Class</b>            | 3                       |
| <b>Packing Group</b>           | II                      |

**IATA**

|                                |                          |
|--------------------------------|--------------------------|
| <b>UN-No</b>                   | UN1224                   |
| <b>Proper Shipping Name</b>    | KETONES, LIQUID, N.O.S.* |
| <b>Technical Shipping Name</b> | (2,3-PENTANEDIONE)       |
| <b>Hazard Class</b>            | 3                        |
| <b>Packing Group</b>           | II                       |

|                                     |                                 |
|-------------------------------------|---------------------------------|
| <b>Special Precautions for User</b> | No special precautions required |
|-------------------------------------|---------------------------------|

**SECTION 15. REGULATORY INFORMATION**

## 2,3-Pentanedione

## International Inventories

China, X = listed, Australia, U.S.A. (TSCA), Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Australia (AICS), Korea (KECL), China (IECSC), Japan (ENCS), Philippines (PICCS), Taiwan (TCSI), Japan (ISHL), New Zealand (NZIoC), Japan (ISHL).

| Component        | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|------------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| 2,3-Pentanedione | -                                                   | -                                       | X    | X     | 209-984-8 | X    | X   | X     | X    | X    | X    | KE-27992 |

## National Regulations

## SECTION 16. OTHER INFORMATION

## Prepared By

Health, Safety and Environmental Department

## Revision Date

27-Apr-2024

## Revision Summary

New emergency telephone response service provider.

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Legend

CAS - Chemical Abstracts Service

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

PICCS - Philippines Inventory of Chemicals and Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

AICS - Australian Inventory of Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit

TWA - Time Weighted Average

ACGIH - American Conference of Governmental Industrial Hygienists

IARC - International Agency for Research on Cancer

DNEL - Derived No Effect Level

PNEC - Predicted No Effect Concentration

RPE - Respiratory Protective Equipment

LD50 - Lethal Dose 50%

LC50 - Lethal Concentration 50%

EC50 - Effective Concentration 50%

NOEC - No Observed Effect Concentration

POW - Partition coefficient Octanol:Water

PBT - Persistent, Bioaccumulative, Toxic

vPvB - very Persistent, very Bioaccumulative

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road

MARPOL - International Convention for the Prevention of Pollution from Ships

OECD - Organisation for Economic Co-operation and Development

ATE - Acute Toxicity Estimate

BCF - Bioconcentration factor

VOC - (Volatile Organic Compound)

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS



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**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**